JACK FARLEY

437·776·9504 ◇ jack@jackfarley.ca ◇ jackfarley.ca

Education

The University of British Columbia **2025–2027 (expected)**
M.Sc. in Statistics *CGPA: 4.33/4.33*
Thesis: A GPU-accelerated parallel-in-time EM algorithm for hidden Markov models with application to cetacean diving behaviour

The University of British Columbia **2020–2025**
B.Sc. Honours in Mathematics *with Distinction* *CGPA: 4.2/4.33*

Research Experience

Department of Statistics *Master's student* **2025–2027**
↪ Profs. Alexandre Bouchard-Côté and Marie Auger-Méthé *UBC*

- Developed a hidden Markov model framework for inferring continuous-time cetacean diving behaviour from aggregated, coarsely sampled data.
- Designed and implemented a parallel-in-time expectation–maximization (EM) algorithm for hidden Markov model parameter estimation in Julia, using custom CUDA kernels.
- Joined a Fisheries and Oceans Canada field camp in the High Arctic for a month, assisting with tagging and biopsying narwhals.

Statistical Consulting Practicum *Statistical consultant* **2026**

- *Follow-up practices after bronchiolitis discharge in Canadian hospitals* BCCHR
- *Comparison of uGFR and eGFR in pediatric kidney transplant patients* BCCHR
- *Statistical approaches for quantifying relative appendage length and Allen's rule* UBC Zoology

Irwin Lab *Undergraduate student researcher* **2023–2025**
↪ Prof. Darren Irwin *UBC*

- Built an individual-based model of hybrid zone formation in Julia, examining how magic traits and search costs affect species persistence after secondary contact between diverging populations.
- Resulting first-author manuscript is under peer review (see Publications).

Ontario Ministry of Natural Resources *Student wildlife biologist* **Summer 2022**
↪ Rod Brook

- Carried out field protocols including nest searching, aerial polar bear surveys, goose banding, and climate monitoring from a remote research station in the subarctic.

Publications

In Review

Farley, J. R., & Irwin, D. (2026). *Magic traits, search costs, and the persistence of species in secondary contact.* bioRxiv. https://doi.org/10.64898/2026.07.23.740407

Non-refereed

Term paper selected for placement in the UBC cIRcle database:

Howard, A., & Farley, J. (2021). *Differences in epiphyte community composition between ornamental and native hardwoods in urban Vancouver [R].* https://doi.org/10.14288/1.0398550

Presentations

International Statistical Ecology Conference **2027**
A parallel-in-time method for parameter estimation in hidden Markov models (poster, upcoming).

Canadian Undergraduate Mathematics Conference **2024**
The mathematics of hybrid zones.

Teaching and Mentorship

Teaching Assistant *STAT 443: Time Series and Forecasting* **2026**

- Led labs, held office hours, and graded assignments and exams.

First Generation Student Mentor *UBC Statistics* **2025–2026**

- Mentored a first-generation undergraduate student, advising on academics and career goals.

Awards and Grants

Polar Knowledge Canada—Northern Scientific Training Program Grant ($3,108) **2026**
British Columbia Graduate Scholarship ($17,500) **2025**
UBC—BPOC Graduate Excellence Award ($3,000) **2025**
UBC—Faculty of Science Graduate Award ($2,000) **2025**
NSERC—Canada Graduate Scholarships – Master's ($27,000, declined) **2025**
UBC—Reginald Palliser-Wilson Scholarship ($3,175) **2024**
UBC—Trek Excellence Scholarship ($1,500) **2023**

Other Professional/Volunteer Experience

Thunder Cape Bird Observatory *Volunteer bird bander* **April–May 2025**

- Banded birds from a remote migration monitoring station.

Long Point Bird Observatory *Volunteer bird bander* **April–May 2024**

- Banded birds from a remote migration monitoring station.

Fundación Maquipucuna & The Sustainable Roots Foundation *Intern* **September 2021 – May 2022**

- Conducted bird surveys, maintained reserves, and guided tourists at research stations in Ecuador.

Delmore Lab, Texas A&M University *Volunteer field technician* **June 2021**

- Recaptured and banded Swainson's Thrushes in northern British Columbia.

Web Development nachiketchildrenslibraries.org **January–March 2021**

- Developed a website for a nonprofit dedicated to building libraries in rural India.

Killarney Provincial Park *Student park ranger* **July–August 2020**

- Maintained park facilities.

James Bay Shorebird Project, Canadian Wildlife Service *Volunteer field technician* **August 2018, 2019**

- Conducted shorebird surveys from a remote field station along the James Bay coast.

Royal Ontario Museum–Ornithological Collections *Volunteer and co-op student* **2017–2020**

- Organized ornithological collections and entered historical data.

Research Interests

I develop hidden Markov model methods for time-series inference, using parallel-in-time algorithms and GPU computation. My current work applies this to modelling whale movement in the Canadian Arctic. I am interested in extending these methods to evolutionary and ecological questions — such as population genetic, phylogenetic, and phylodynamic inference — using Bayesian approaches.

Technical Skills

Programming Julia, R, Java, LaTeX, & CUDA/GPU computing
Statistical Methods MCMC, SMC, & hidden Markov models

Selected Graduate-Level Coursework

Mathematics
- Measure Theory
- Continuous Time Stochastic Processes
- Discrete Random Processes
- Functional Analysis

Statistics
- Statistical Computing
- Spatial Statistics
- Statistical Theory I & II
- Techniques of Statistical Consulting